

# Edge Geography is XNLP-hard for Pathwidth and in XP for Tree-Partition Width

**Thobias Kvalvik Høivik (presenting) · Erlend Raa Vågset**

Western Norway University of Applied Sciences

ESA 2026 · L'Aquila

# Edge Geography

---

One of the most famous PSPACE-complete problems.

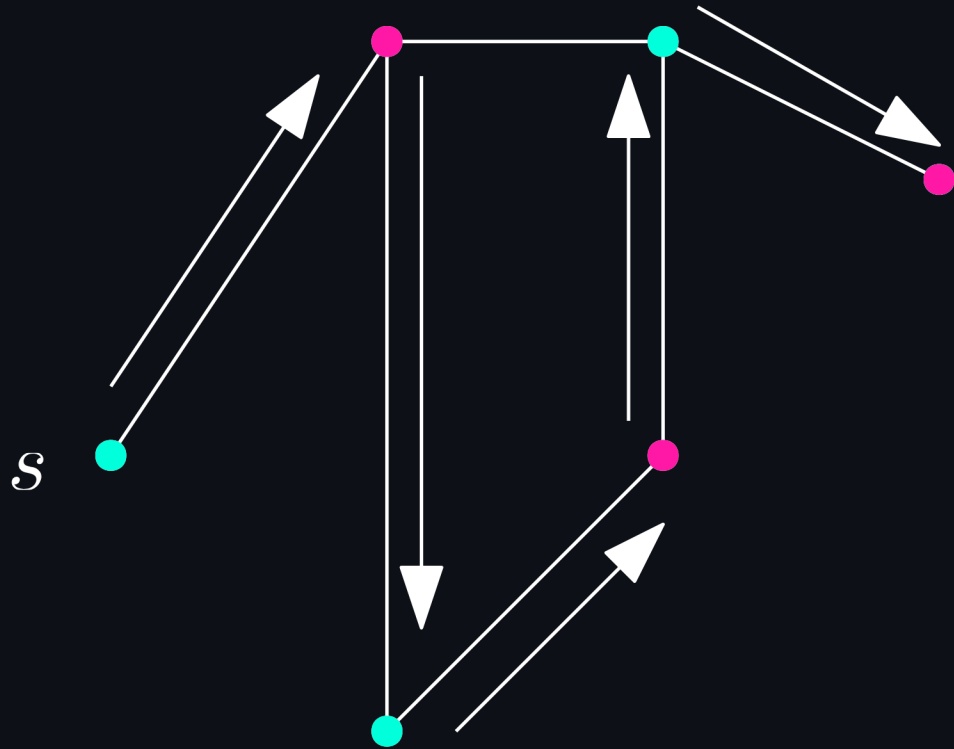
A token starts at a vertex  $s$ .

Players alternate:

1. choose an unused edge from the current vertex,
2. move the token along it,
3. delete that edge.

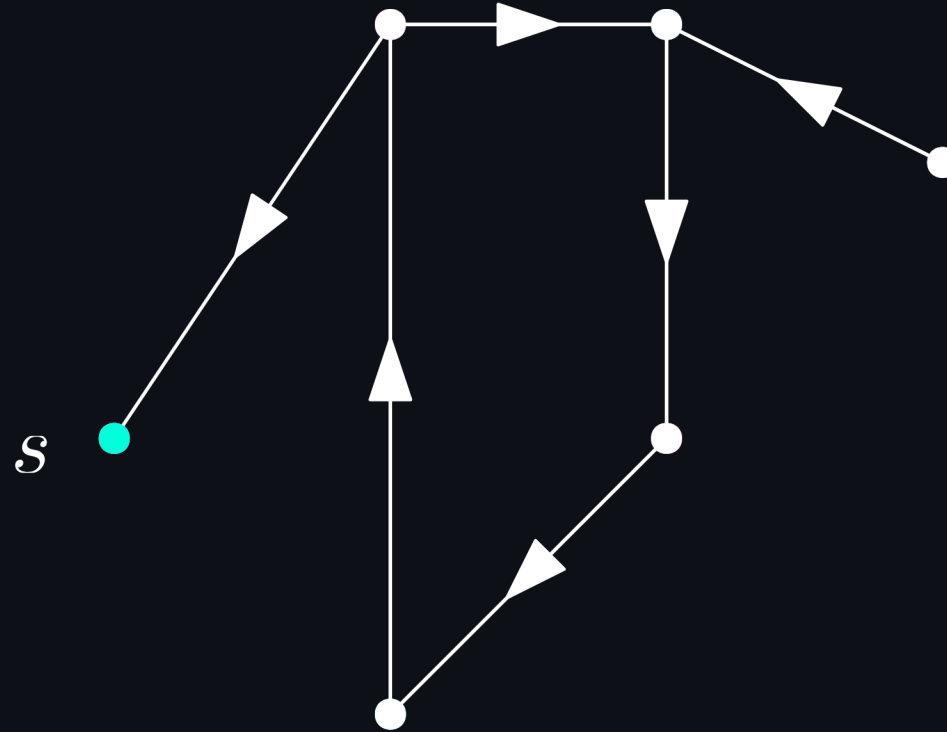
The first player unable to move loses.

## Undirected Edge Geography



Player 1 wins!

## Directed Edge Geography



Player 2 wins!

# Parameterizing the problem

---

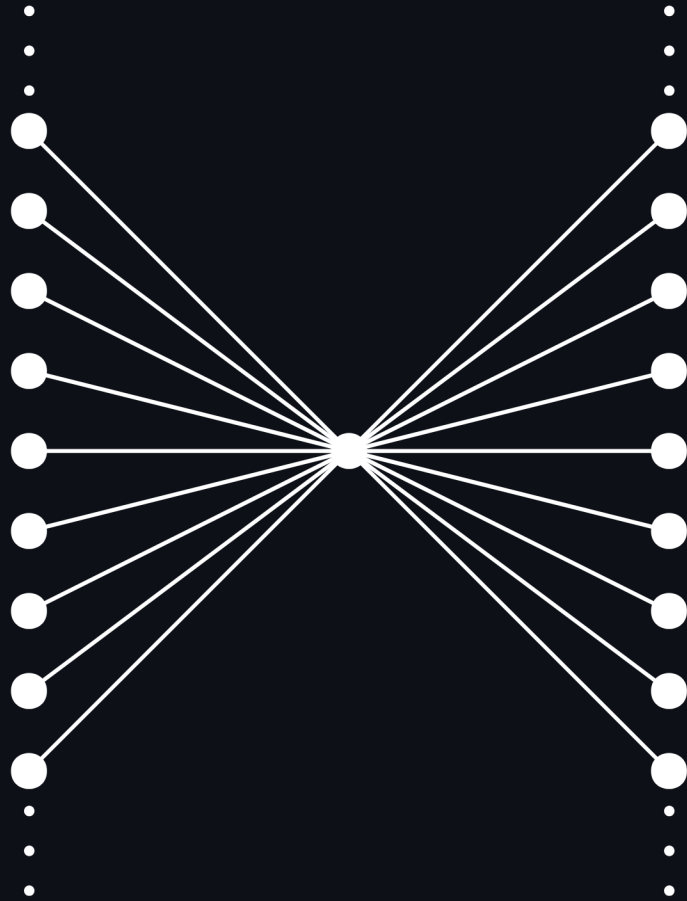
## Theorem (Bodlaender, modern language)

(Undirected) Edge Geography is solvable in time  $f(k, d) \cdot O(n)$  where  $k$  is the treewidth of the graph and  $d$  is the maximum degree of any vertex in the graph.

Exponential time becomes "linear time" when two parameters are controlled!

What if we allow  $d$  to be unbounded?

Arbitrarily many crossings



when  $\Delta(G)$  is unbounded!

# Our first result

---

## Theorem

Directed Edge Geography is XNLP-hard parameterized by pathwidth.

XNLP "rules out" fixed-parameter tractability.

Bounded pathwidth  $\Rightarrow$  bounded treewidth.

Pathwidth, and thus treewidth, is not enough!

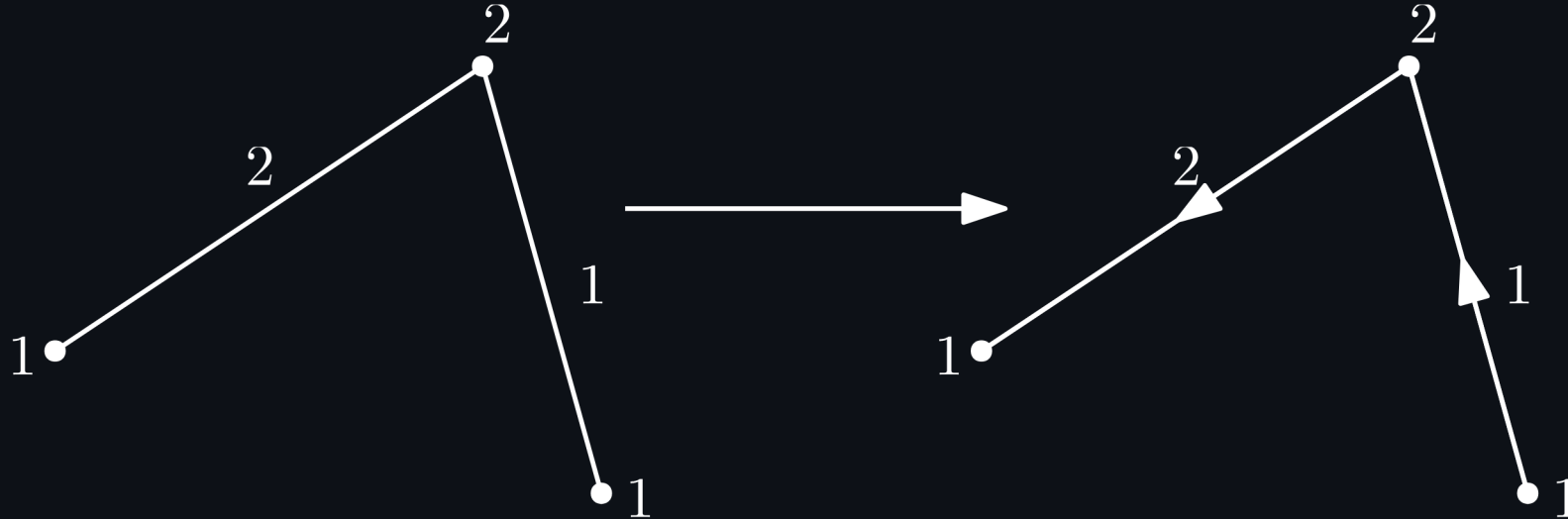
This result transfers to Undirected Edge Geography in a simple way!

# Source problem: Chosen Maximum Outdegree

Does there exist an orientation of every edge so that each vertex respects its budget.

$$\sum_{e \in \theta_w(v)} w(e) \leq t(v) \quad \forall v$$

FIX THIS IMAGE IT IS WRONG



Known XNLP-complete when parameterized by pathwidth.

# Reduction idea in two phases

---

## Phase I — Player 1 encodes an orientation

For each input edge  $e = \{x, y\}$ , Player 1 chooses one of two sides.

$$\text{choose the } y\text{-side} \iff x \rightarrow y$$

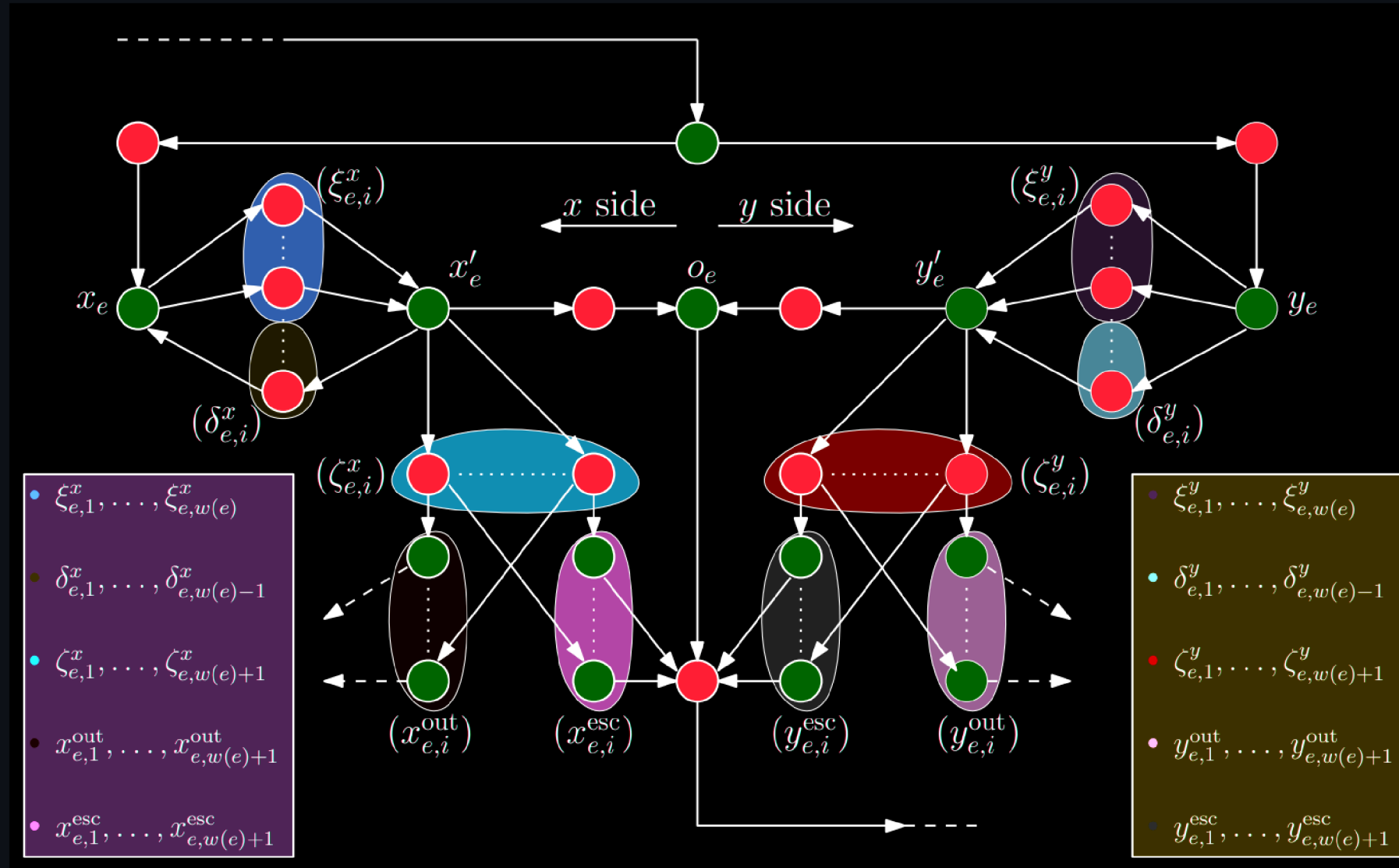
## Phase II — Player 2 challenges a vertex

Player 2 chooses  $x$  and tests the total weight oriented out of  $x$ .

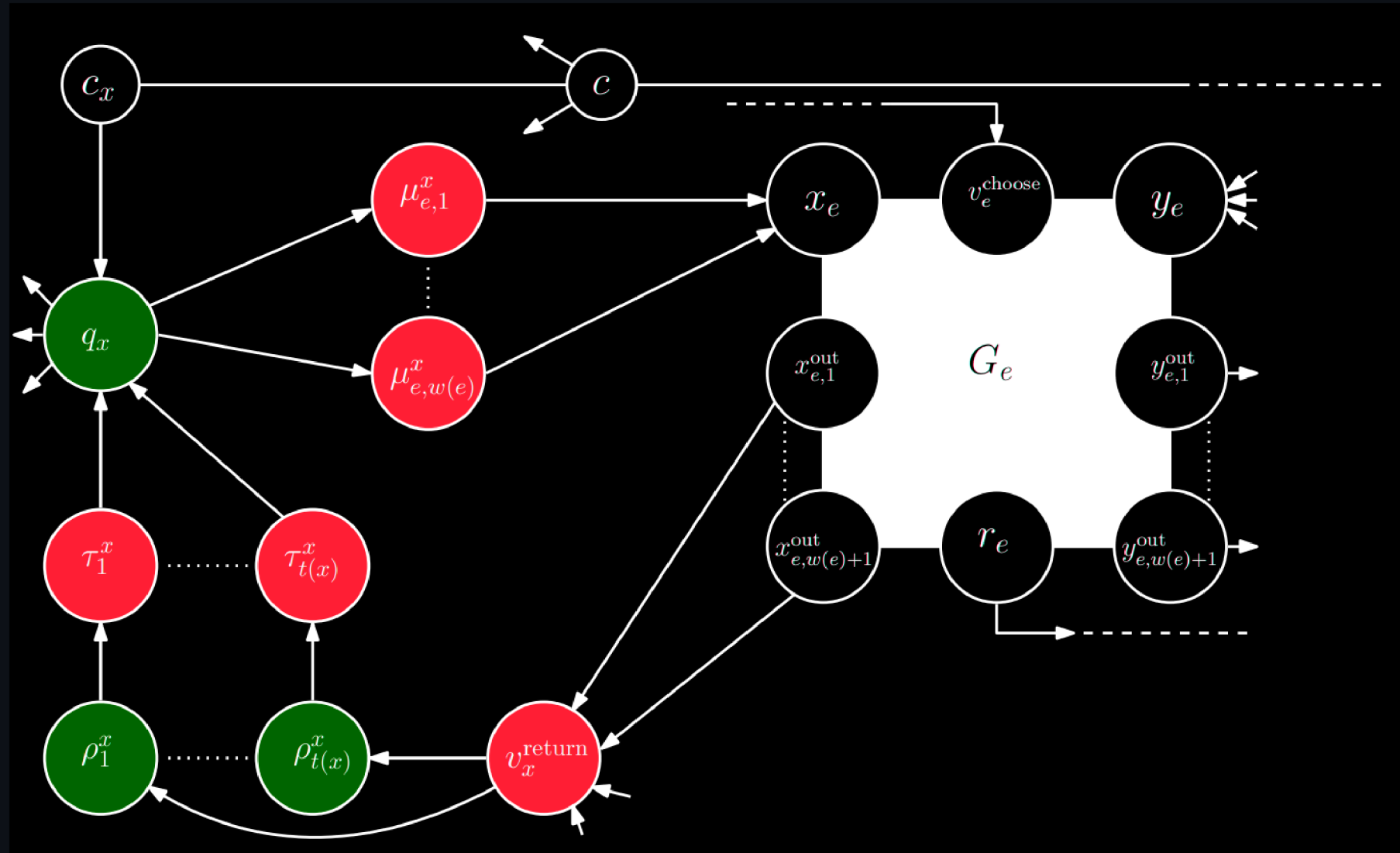
If outgoing weight  $> t(x)$ , Player 2 has one excursion too many.



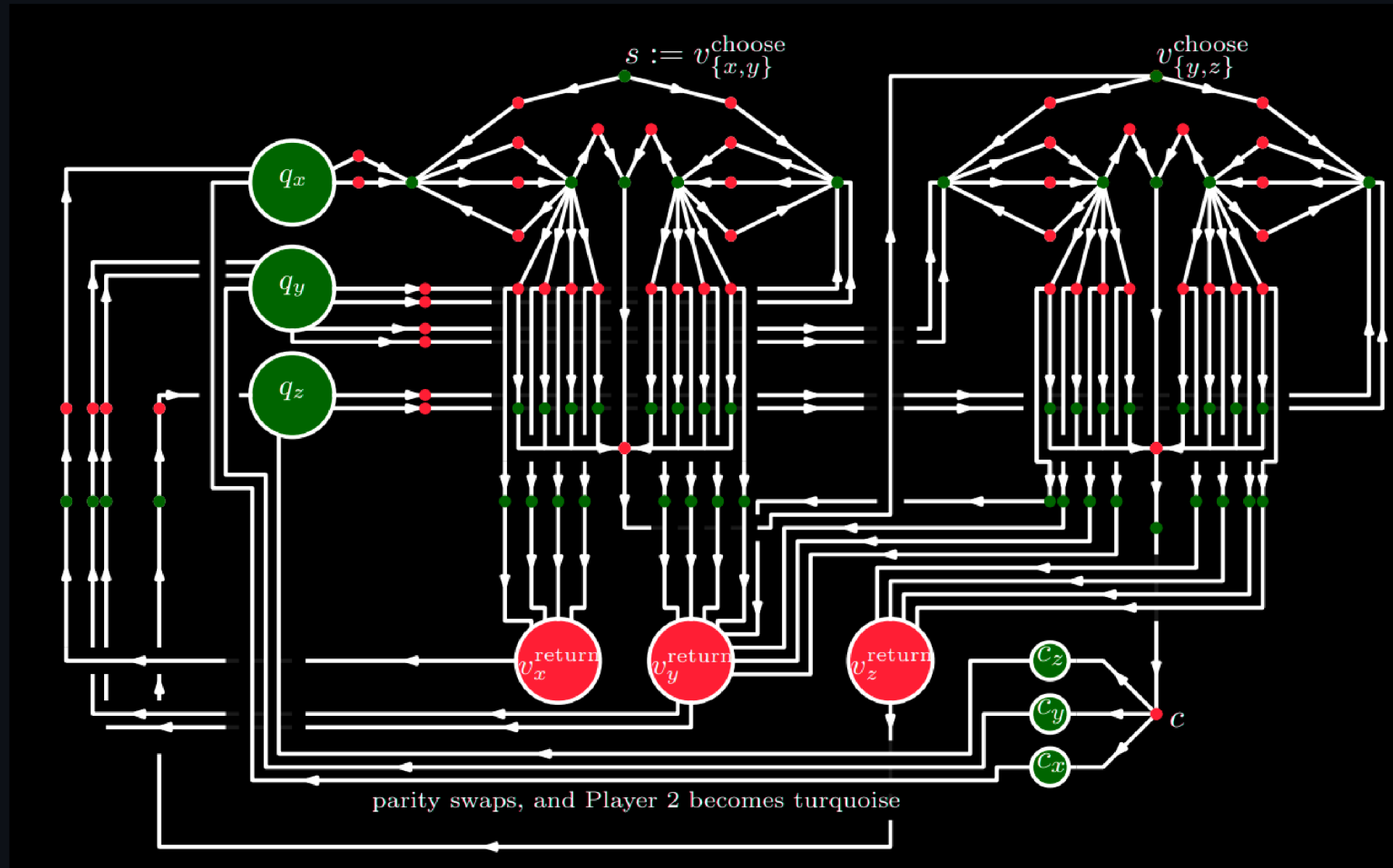
# Phase 1



# Phase 2



# The full reduction for a small source instance



# Why correctness follows

---

## Feasible orientation

$$\sum_{e \in \theta_\omega(x)} w(e) \leq t(x) \text{ for every } x.$$

Player 2 eventually runs out of non-losing challenges.

**Player 1 wins**

## Infeasible orientation

For some  $x$ ,

$$\sum_{e \in \theta_\omega(x)} w(e) > t(x).$$

Player 2 chooses this  $x$  and survives one challenge too many.

**Player 2 wins**

## Why pathwidth is preserved

---

Process an input edge  $e = \{x, y\}$  when  $x$  and  $y$  coexist in the source path decomposition.

Keep only the necessary anchors alive.

Introduce auxiliary gadget vertices locally, then forget them immediately.

$$\text{pw}(G') = O(k)$$

# Directed to undirected

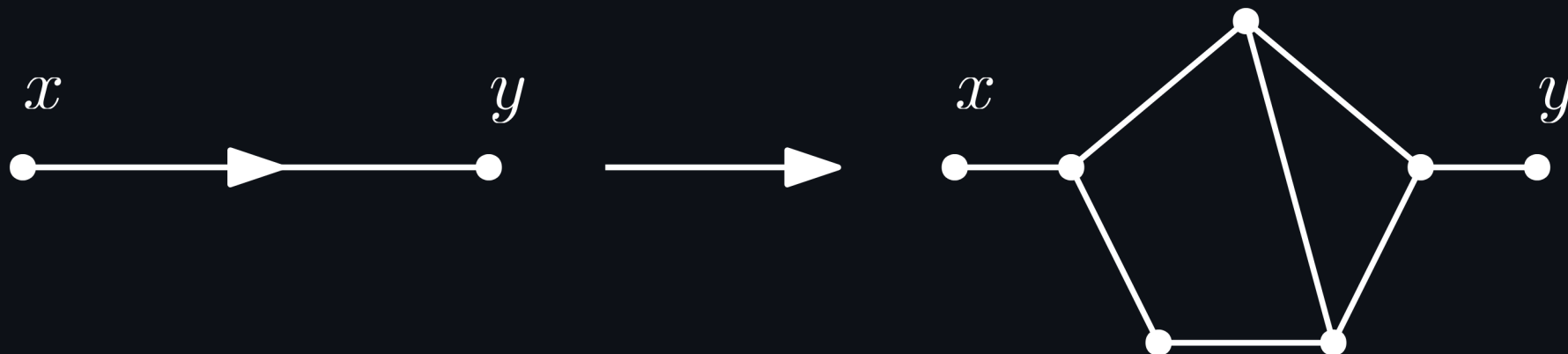
Each directed edge is replaced by a constant-size pseudoarc.

$$u \rightarrow v \quad \rightsquigarrow \quad \text{undirected pseudoarc}$$

Safe traversal works from  $u$  to  $v$ .

Entering the wrong way is losing under optimal play.

Directed hardness transfers to Undirected Edge Geography.



# XP membership for width of treepartition

---

A rooted tree partition groups vertices into bags.

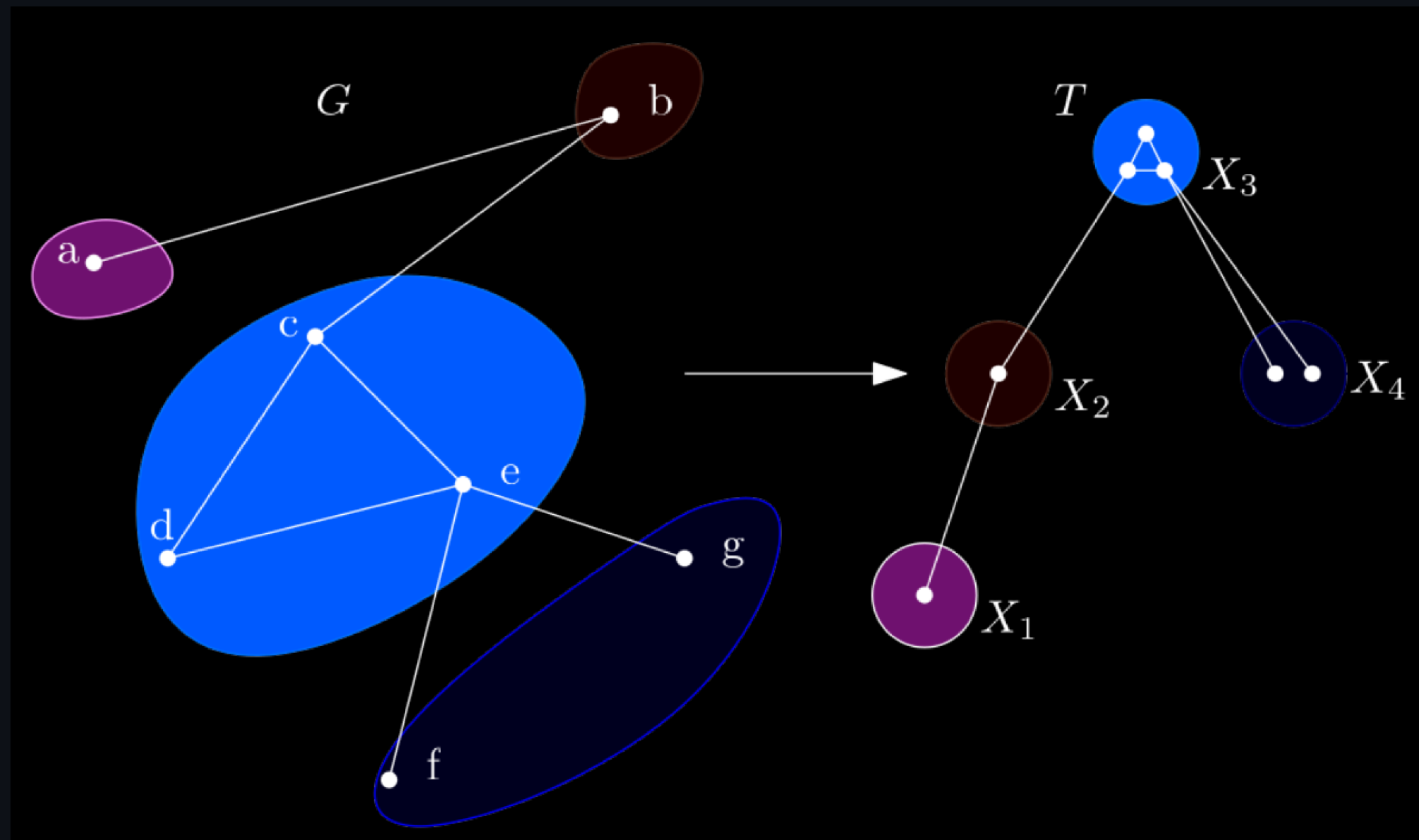
Edges are either:

- inside one bag, or
- between adjacent bags of the tree.

For width  $k$ , a parent-child cut has at most  $k^2$  ports (on simple graphs).

## Theorem

Undirected Edge Geography on simple graphs, given together with a rooted tree partition of width  $k$ , is solvable in XP time parameterized by  $k$ .





# Bottom-up symbolic minimax

leaf  $\longrightarrow$  interface type  $\longrightarrow$  parent  $\longrightarrow \dots \longrightarrow$  root

## 1 · Leaves

For every entry port and local residual state, brute-force minimax **inside the bag**.

Treat possible returns to the parent as variables  $\nu_p$ .

## 2 · Internal bags

Replace each solved child by its interface type  $\sigma$ .

Collect equal types into counters  $m_\sigma$  and run local minimax.

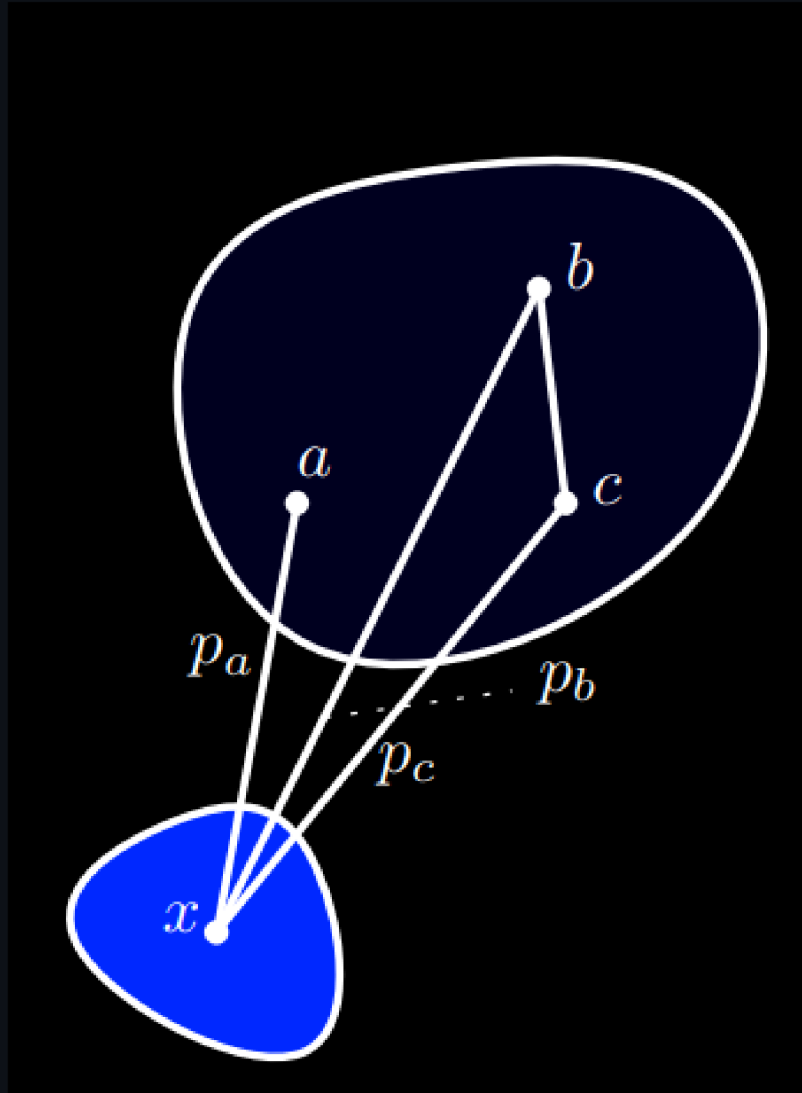
## 3 · Root

There is no unknown continuation above the root.

Evaluate the final compressed state to obtain the winner.

Entering a solved child is a **table lookup**; returning is a **counter update**.

# A solved child conveys a boolean function



The child exports a response

$$\Phi_a(\nu_b, \nu_c) = \nu_b \wedge \nu_c.$$

$\nu_b$  and  $\nu_c$  are the values of the two possible continuations in the parent.

The cut has  $O(k^2)$  ports, so the truth table of  $\Phi_a$  has size only  $f(k)$ .

Returning through  $p_b$  also exports the smaller residual type  $\sigma_c$ .

# Child counters comprise the worst factor

---

## The $n$ -dependent part

For fixed  $k$ , there are  $t_k$  possible child types.

Each counter lies in  $\{0, \dots, n\}$ , so

$$m \in \{0, \dots, n\}^{t_k} \implies \#m \leq (n+1)^{t_k} = n^{O(t_k)}.$$

## The other factors

Unused ports and internal edges contribute

$$2^{k^2} \cdot 2^{\binom{k}{2}} = 2^{O(k^2)}.$$

All other boundary data depends only on  $k$ ; collect it into  $g(k)$ .

$$\underbrace{n}_{\text{bags}} \cdot \underbrace{g(k)}_{\text{includes } 2^{O(k^2)}} \cdot \underbrace{n^{O(t_k)}}_{\text{counter states}} = n^{f(k)}.$$

# Laughably bad in practice

Let  $T_r$  be the number of ambient types on an interface with  $r$  ports.

| ports $r$           | 0 | 1 | 2   | 3                 | 4                   |
|---------------------|---|---|-----|-------------------|---------------------|
| ambient types $T_r$ | 1 | 2 | 676 | $\approx 2^{768}$ | $\geq 2^{2^{4058}}$ |

Illustrative ambient count obtained by unwinding Definition 35.

The number of realizable types may be vastly smaller.

A width- $k$  cut may have  $k^2$  ports.

The crude bound is a tower of exponentials of height  $O(k^2)$ .

# Takeaway

---

## Pathwidth

small separators

**but residual history is  
uncontrolled**

**XNLP-hard**

## Tree-partition width

bounded interfaces

**residual play can be compressed**

**XP**

Open: Is Edge Geography in XP parameterized by pathwidth alone?